# Subgroup Performance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

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## Subgroup Performance

> Prepare Subgroup Performance: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Calculate **subgroup** results with sample sizes

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

### Improve Generalization

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

Look for: Compare **training** and validation **evidence** for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

### Learning Curves

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

Look for: Read **training**-size learning curves

Reference example, not the only acceptable wording: Teacher explanation: Mean(5+7+6)/3=6; range 7-5=2.

### Metrics Match Consequences

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a declared score threshold including equality

Reference example, not the only acceptable wording: Teacher explanation: At .5: A, B, C. At .3: A, B, C, D, E, including equality.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

> Device A gets 18/20 correct; B gets 6/10. Compute overall accuracy and the **subgroup** gap. What causal claim is not supported?

### Concept to explain

Overall metrics can conceal uneven behavior across meaningful conditions or user groups.

| Term | Meaning |
| --- | --- |
| subgroup | meaningful slice of evaluation cases |
| performance gap | difference between subgroup metrics |
| denominator | number of cases behind a rate |
| intersection | cases belonging to multiple slices |

### Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

## Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on **subgroup** evaluation. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

### Readiness answer

Different error patterns in different evaluated conditions or groups.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

### End goal for the pictorial case

By the end, I can explain **subgroup** performance using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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## Explain the mechanism

> Model Subgroup Performance using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

### Question to explain

QUESTION Calculate **subgroup** accuracy and recall gaps with adequate **denominator**s.

MODEL IDEA Overall metrics can conceal uneven behavior across meaningful conditions or user groups.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

### How the mechanism works

UNDER THE HOOD Daylight: 18/20 correct = 90%; evening: 10/20 = 50%. Accuracy gap = 90% – 50% = 40 percentage points.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

### Step 1

STEP 1 Daylight: 18/20 correct = 90%; evening: 10/20 = 50%.

### Step 2

STEP 2 Accuracy gap = 90% – 50% = 40 percentage points.

### Step 3

STEP 3 Evening spotted recall is 4/10 = 40%, versus daylight 8/10 = **80%**.

### Step 4

STEP 4 Inspect representation and image quality; report counts and uncertainty before repairing.

CHECK Verify the **denominator**, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

**Concrete teaching case**

### The teaching case

A has18/20 correct; B has6/10.

### The teaching case

Compute **subgroup** results and **evidence** limits.

**Exact explanation to model**

### Why the result follows

90% versus**60%**, gap30 percentage points. Report **denominator**s and investigate collection/label differences without assuming an inherent group cause.

**Explain the added worked example**

## Explain the added worked example · 1

A single overall score can hide different experiences. Define relevant groups before evaluation and report counts as well as rates. This classroom exercise uses fictional sensor conditions, not personal identities.

Real evaluations involving people require appropriate permissions, privacy and contextual expertise.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **subgroup** evaluation.

Sensor group A has 2 errors in 20 examples and group B has 3 in 5. Error rates are 10 and 60 percent; pooled error is 5/25=20 percent. The group difference is 50 percentage points.

B has only five examples, so its rate is sensitive to each individual result.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

### Key terms

| Term | Meaning |
| --- | --- |
| Subgroup | A defined subset of the evaluated examples |
| Disaggregation | Reporting results separately for defined groups |
| Coverage | Which groups and conditions are represented |
| Disparity | An observed difference between group results |

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

### Subgroup evaluation at a glance

| PART | EVIDENCE OR ACTION | INTERPRETATION |
| --- | --- | --- |
| Group A | 2 errors of 20 | 10 percent |
| Group B | 3 errors of 5 | 60 percent |
| Pooled | 5 errors of 25 | 20 percent |

Changing one B error to a correct result changes its rate from 60 to 40 percent. Small samples can be unstable without making the observed errors unimportant. A missing group has no measured rate, not zero error. Intersections of conditions may be even smaller.

**How does correcting one B error affect its rate? Explain the steps or evidence behind your response.**

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Subgroup Performance

Calculate **subgroup** and pooled error rates without letting the larger group hide a gap.

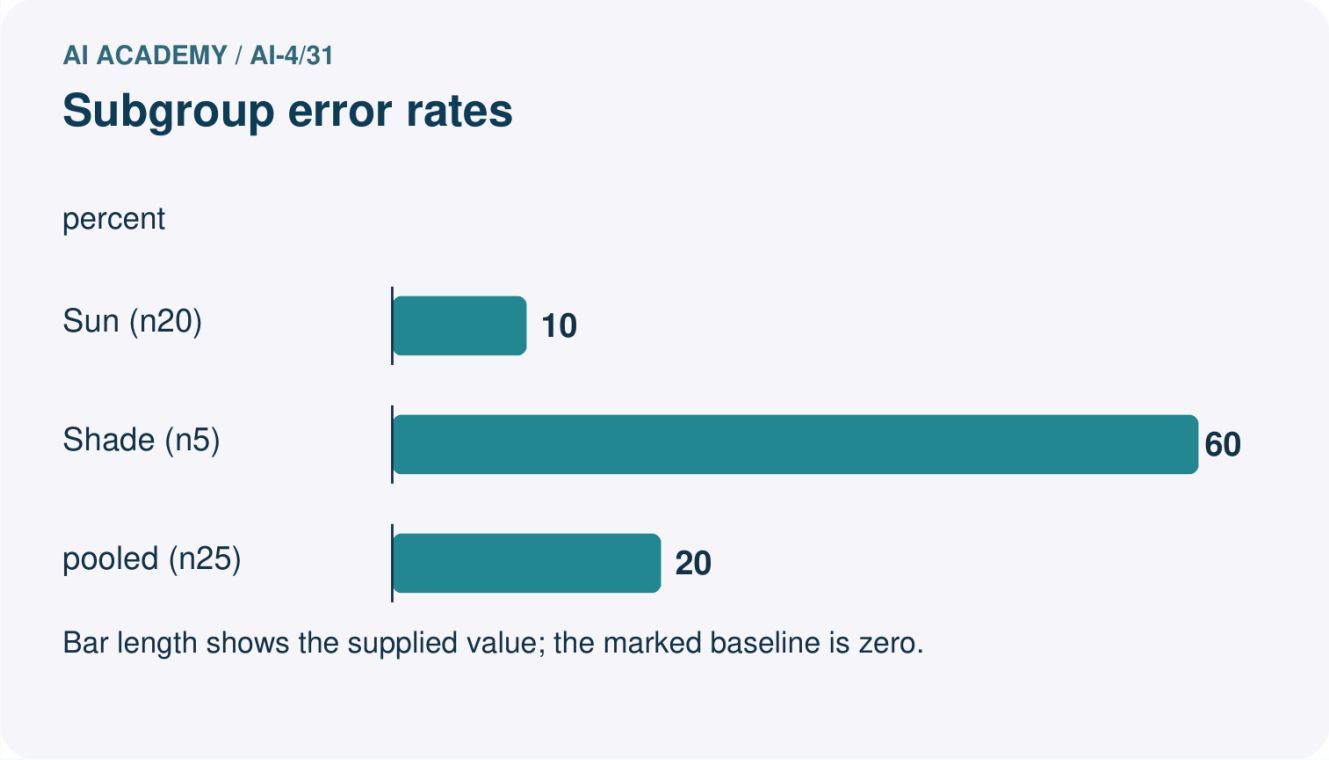
### The facts we will use

#### Supplied facts

Group Sun: 2 errors in20 photos; Shade: 3 errors in5. Same error definition; related plant views are documented.

#### Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.




Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/31

# Subgroup Performance

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Sun rate

Sun error=2/20=10%.





### 2. Shade rate

Shade error=3/5=60%.





### 3. Pooled rate

Total error=5/25=20%, which hides the 50-percentage-point subgroup difference.

**CHANGE ONE THING**

Shade gains5 new cases with no errors.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Sun rate | Sun error=2/20=10%. |
| 2. Shade rate | Shade error=3/5=60%. |
| 3. Pooled rate | Total error=5/25=20%, which hides the 50-percentage-point subgroup difference. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

Read the picture in order. Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

#### Keep the stated limit

Small **subgroup** samples have high variability. An observed gap does not by itself establish

discrimination or a causal explanation.

**Change one thing in the pictured case**

### Change one thing

Change: Shade gains5 new cases with no errors.

### Predicted result

**Prediction**: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%.

### Why the result follows

Reason: Extra evaluated cases change **denominator**s; existing errors have not disappeared.

**Ask while showing the picture**

### Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

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## Demonstrate and guide practice

> Demonstrate Subgroup Performance, reduce support deliberately and check individual reasoning before discussing answers.

### Choose the pacing route

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

**Board sequence**

### 1. Put the facts on the board

Write the given facts: A has18/20 correct; B has6/10.

### 2. State the decision to explain

Write the question to explain: Compute **subgroup** results and **evidence** limits.

### 3. Reveal the worked reasoning

Reveal after **prediction**s: 90% versus**60%**, gap30 percentage points. Report **denominator**s and investigate collection/label differences without assuming an inherent group cause.

### Hold back the independent task

Keep for an independent attempt: Accuracy is 18/20 in bright light and 6/10 in dim light. Compute both rates and the percentage-point gap.

### Classroom dialogue

#### Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

#### Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and **denominator**s for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### Model these guided questions

Guided response: B becomes 2/5=40 percent, a 20-percentage-point change from one case. Exit response: Error types, consequences, access, labels and context can differ; a single equality does not establish acceptable treatment.

### Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: What can a pooled error rate hide? Check the explanation before proceeding.

> **Stage:** 7 to 18 minutes

**Teacher action:** Read the opening and model the worked evidence. Ask students to identify subgroup in context.

**Stage:** 18 to 32 minutes

**Teacher action:** Complete W 1 and W 2 with support: calculate group rates; check sensitivity. Require the evidence, not only the final response.

**Stage:** 32 to 45 minutes

**Teacher action:** Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

**Stage:** Session 2 start to 8 minutes

**Teacher action:** Retrieve the distinction between subgroup and disaggregation. Invite a counterexample to an overconfident claim.

**Stage:** 8 to 20 minutes

**Teacher action:** Discuss the application and complete W 3: Group C has no evaluated cases. What is its measured error rate?

**Stage:** 20 to 35 minutes

**Teacher action:** Collect W4 independently before giving hints. Assess these outcomes: Calculate subgroup results with sample sizes; Distinguish missing evidence from equal performance. Record any support used.

**Stage:** 35 to 45 minutes

**Teacher action:** Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

### Reduce help deliberately

#### Completion practice

Completion task: How does correcting one B error affect its rate?

#### Reasoning to listen for

Teacher check: Teacher explanation: B becomes 2/5=40 percent, a 20-percentage-point change from one case.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for **Subgroup** evaluation. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

#### Expected repair

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: A 2/20=10 percent; B 3/5=60 percent; pooled 5/25=20 percent, a 50-point group gap.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### Pictorial practice question

Why should the report retain **subgroup** rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

### Reasoned teacher answer

The20% original total is dominated by Sun’s larger sample and understates the observed Shade difficulty. Include counts, conditions and uncertainty.

Changed-case check: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Extra evaluated cases change **denominator**s; existing errors have not disappeared.

Changed-case reasoning: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Extra evaluated cases change **denominator**s; existing errors have not disappeared.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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TEACHER GUIDE · 4 OF 9

## Run the weekly evidence lab

> Run the Subgroup Performance investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Slice Audit: compute metrics and **denominator**s for lighting, species, and camera; flag small samples and prioritize one gap.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

### Diagnose and reteach

MISCONCEPTION TO REPAIR A good overall result guarantees acceptable performance for every **subgroup**.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: A 2/20=10 percent; B 3/5=60 percent; pooled 5/25=20 percent, a 50-point group gap.

A stronger claim makes a stronger project: Error types, consequences, access, labels and context can differ; a single equality does not establish acceptable treatment.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: X 10 percent; Y 50 percent; pooled 3/14 about 21.4 percent.

Y has only four examples, and the table does not establish a cause.

### Match the support to the first error

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

## Assess and explain the answers

> Use the Subgroup Performance questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Worked case answer

#### Teacher key · Worked case answer

90% versus**60%**, gap30 percentage points. Report **denominator**s and investigate collection/label differences without assuming an inherent group cause.

### Independent question

#### Independent question · withhold the key

Accuracy is 18/20 in bright light and 6/10 in dim light. Compute both rates and the percentage-point gap.

### Expected independent answer

#### Teacher key · Expected independent answer

Bright: 90%; dim: **60%**; gap: 30 percentage points. Report **denominator**s and uncertainty from small samples before making broad claims.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: “The difference is 30 percent of all examples rather than 30 percentage points between rates.” Keep the first attempt and an uncoached retry separate.

### Original transfer answer

#### Teacher key · Original transfer answer

Overall=24/30=**80%**; A=90%, B=**60%**, gap=30 percentage points. These data do not prove the device itself causes the difference; lighting, leaf mix or labels may differ.

Investigate with comparable cases and preserve small-sample uncertainty.

### Explanations for the additional workbook examples

### W1 Calculate group rates

#### Workbook question · W1

Calculate the worked group and pooled rates.

### Teacher reasoning · W1

Teacher explanation: A 2/20=10 percent; B 3/5=60 percent; pooled 5/25=20 percent, a 50-point group gap.

**W2 Check sensitivity**

### Workbook question · W2

How does correcting one B error affect its rate?

### Teacher reasoning · W2

Teacher explanation: B becomes 2/5=40 percent, a 20-percentage-point change from one case.

**W3 Report missing coverage**

### Workbook question · W3

Group C has no evaluated cases. What is its measured error rate?

### Teacher reasoning · W3

Teacher explanation: Unavailable because the **denominator** is zero; report missing **coverage** and do not substitute zero.

**W4 Apply independently**

### Workbook question · W4

Groups X 1 error/10 and Y 2/4: calculate separate and pooled rates, then state one limit.

### Teacher reasoning · W4

Teacher explanation: X 10 percent; Y 50 percent; pooled 3/14 about 21.4 percent. Y has only four examples, and the table does not establish a cause.

**W5 Avoid a fairness shortcut**

### Workbook question · W5

Why does equal accuracy not settle whether use is fair?

### Teacher reasoning · W5

Teacher explanation: Error types, consequences, access, labels and context can differ; a single equality does not establish acceptable treatment.

**Assess independent transfer**

### Independent question · withhold the key

Selected case: Groups X 1 error/10 and Y 2/4: calculate separate and pooled rates, then state one limit.

**Selected case answer**

### Teacher key · Selected case answer

Teacher explanation: X 10 percent; Y 50 percent; pooled 3/14 about 21.4 percent. Y has only four examples, and the table does not establish a cause.

| Dimension | Evidence required |
| --- | --- |
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect and extend learning

> Connect Subgroup Performance to Responsible Plant Classifier, adapt support and plan a later changed-case check.

### Responsible Plant Classifier

Add **subgroup** tables and a minimum acceptable threshold to the Plant Classifier release criteria.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

### Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and **subgroup** or shifted-condition analysis with **denominator**s.

### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

### Bridge to the next week

Next: Week 32: Proxy Features and Fairness. Retrieve today's **evidence** and identify what carries forward and what is new.

### Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **subgroup** and **disaggregation** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: X 10 percent; Y 50 percent; pooled 3/14 about 21.4 percent. Y has only four examples, and the table does not establish a cause.

Fresh retry X 2/8 and Y 1/2:25 and 50 percent, pooled 3/10=30 percent. Require **denominator**s, the small Y sample and no causal inference. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Proxy features and fairness. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

### Plan the delayed return

Suggested return points in this level: ai-4/32, ai-4/34. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
|---|---|
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

# Prepare, teach and assess

> Use this preparation summary alongside the detailed Subgroup Performance lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Calculate **subgroup** results with sample sizes

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: How does correcting one B error affect its rate?

### Supported practice answer

Reasoned practice answer: Teacher explanation: B becomes 2/5=40 percent, a 20-percentage-point change from one case.

Check every learner before discussion: Use the worked case for **Subgroup** evaluation. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: A 2/20=10 percent; B 3/5=60 percent; pooled 5/25=20 percent, a 50-point group gap.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: Shade gains5 new cases with no errors. Require a **prediction** and a reason before comparing. Expected result: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%.

Reason: Extra evaluated cases change **denominator**s; existing errors have not disappeared.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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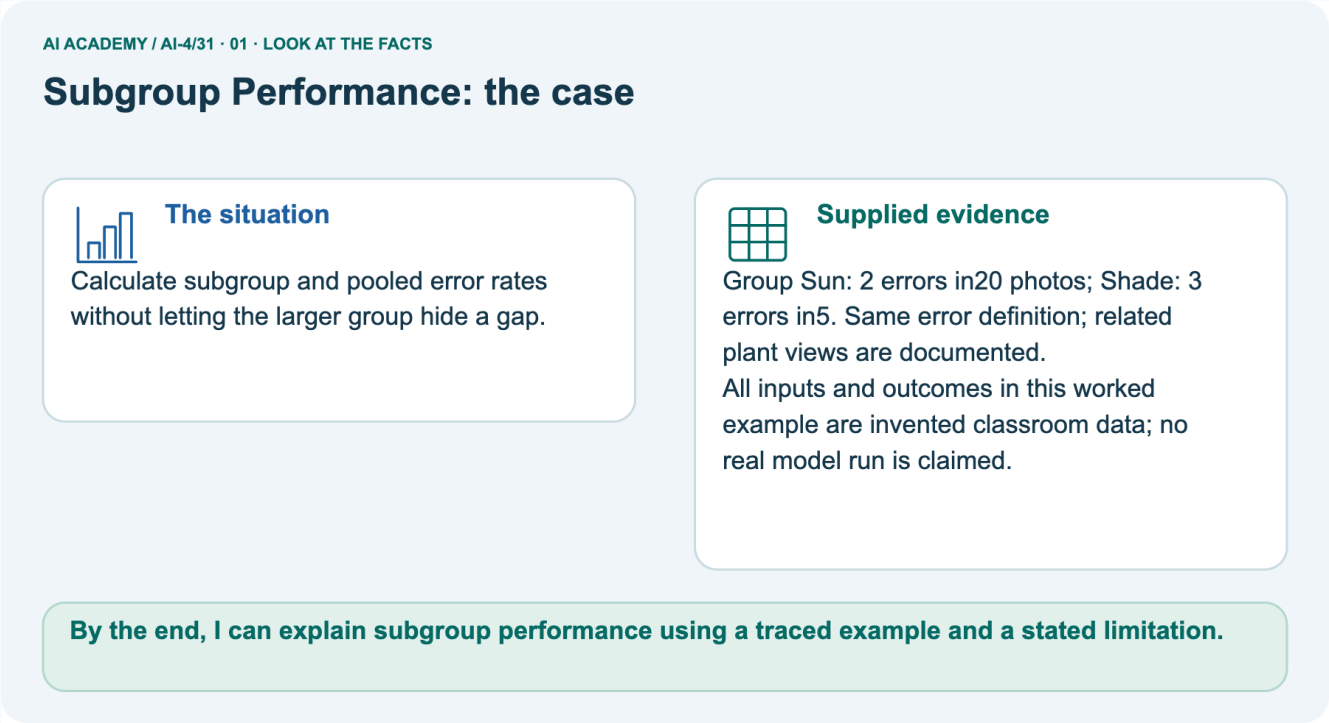


## Visual concept journey

> Show one Subgroup Performance graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

### 01. Subgroup Performance: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate **subgroup** and pooled error rates without letting the larger group hide a gap. Supplied facts: Group Sun: 2 errors in20 photos; Shade: 3 errors in5. Same error definition; related plant views are documented.

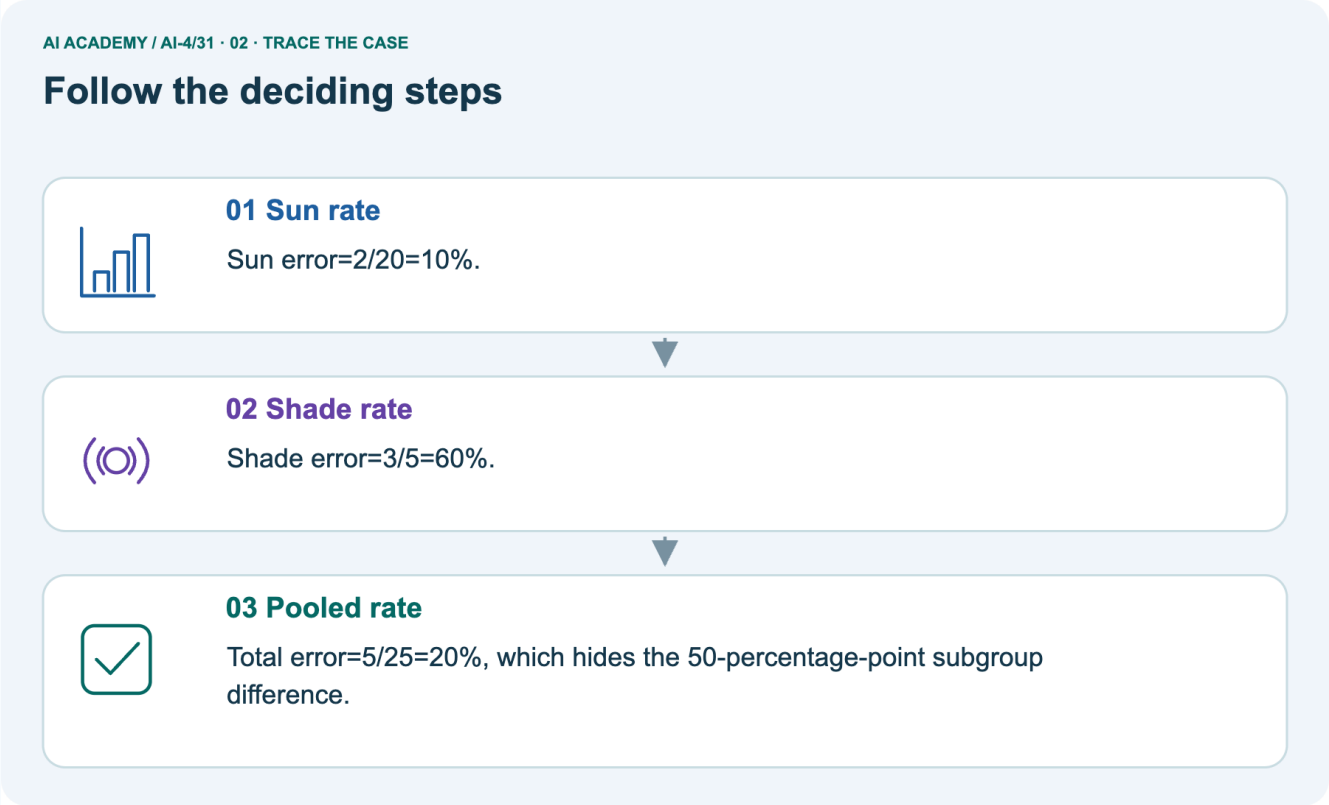
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

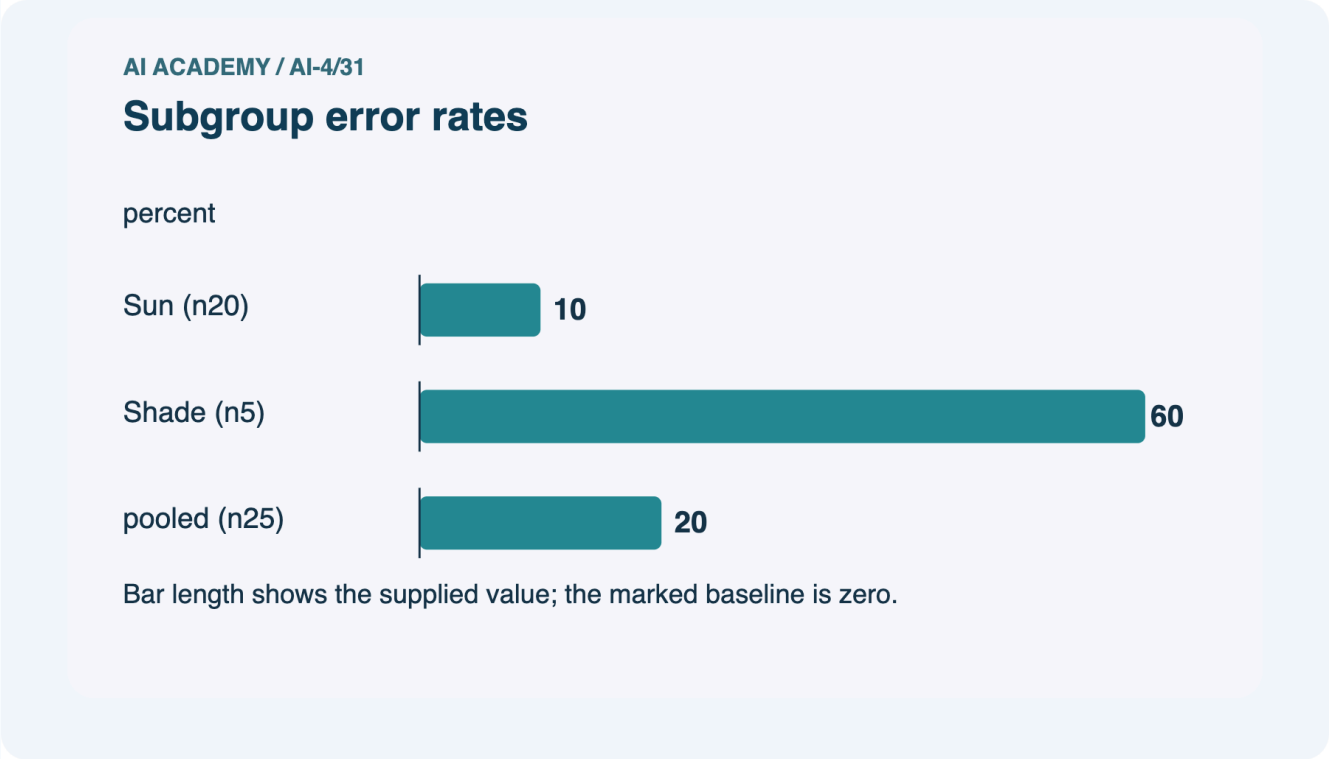
Sun rate: Sun error=2/20=10%. → Shade rate: Shade error=3/5=**60%**. → Pooled rate: Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference.

Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 03. Subgroup error rates




Follow the labelled picture with the matching explanation. Open the vector diagram

**Subgroup** error rates. Units: percent. Sun (n20): 10. Shade (n5): 60. pooled (n25): 20.

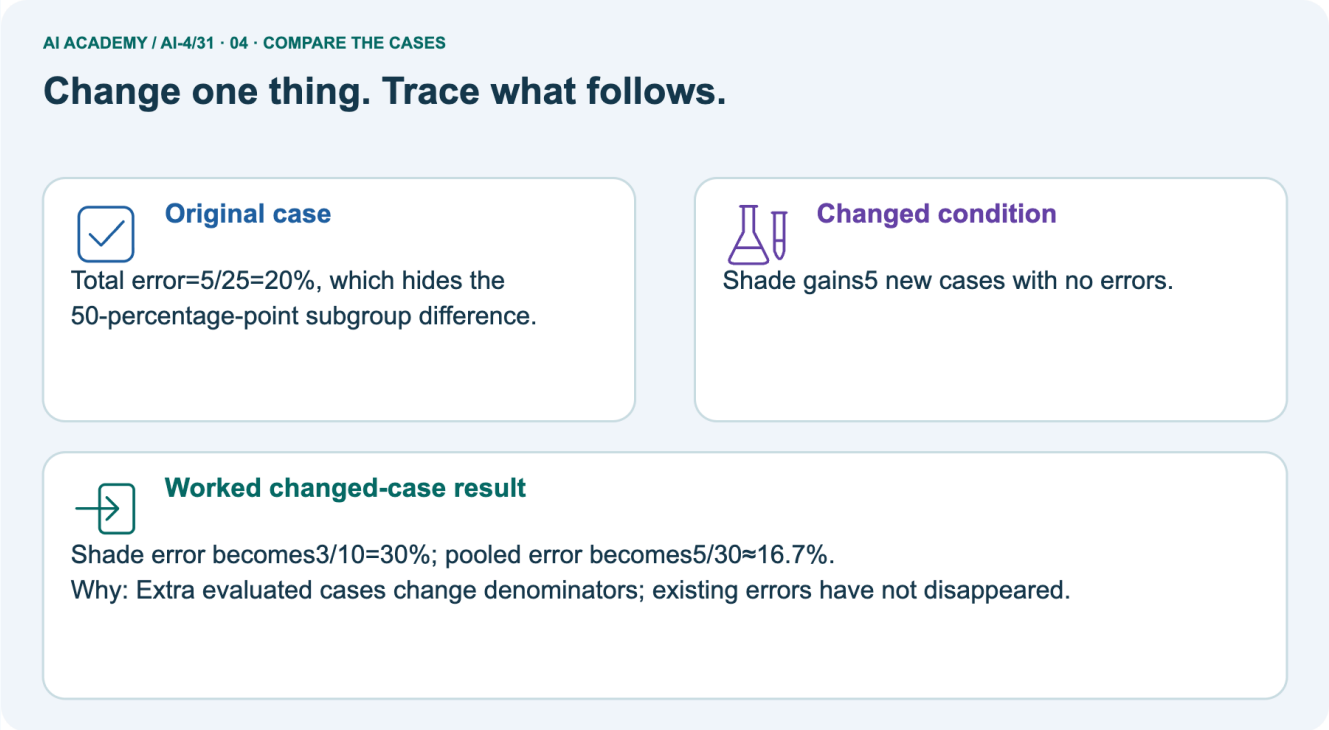
Read the labels and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and **denominator**s before discussing a difference. Supplied toy values are not real class or model results.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. Changed condition: Shade gains5 new cases with no errors.

Expected result: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Reason: Extra evaluated cases change **denominator**s; existing errors have not disappeared.

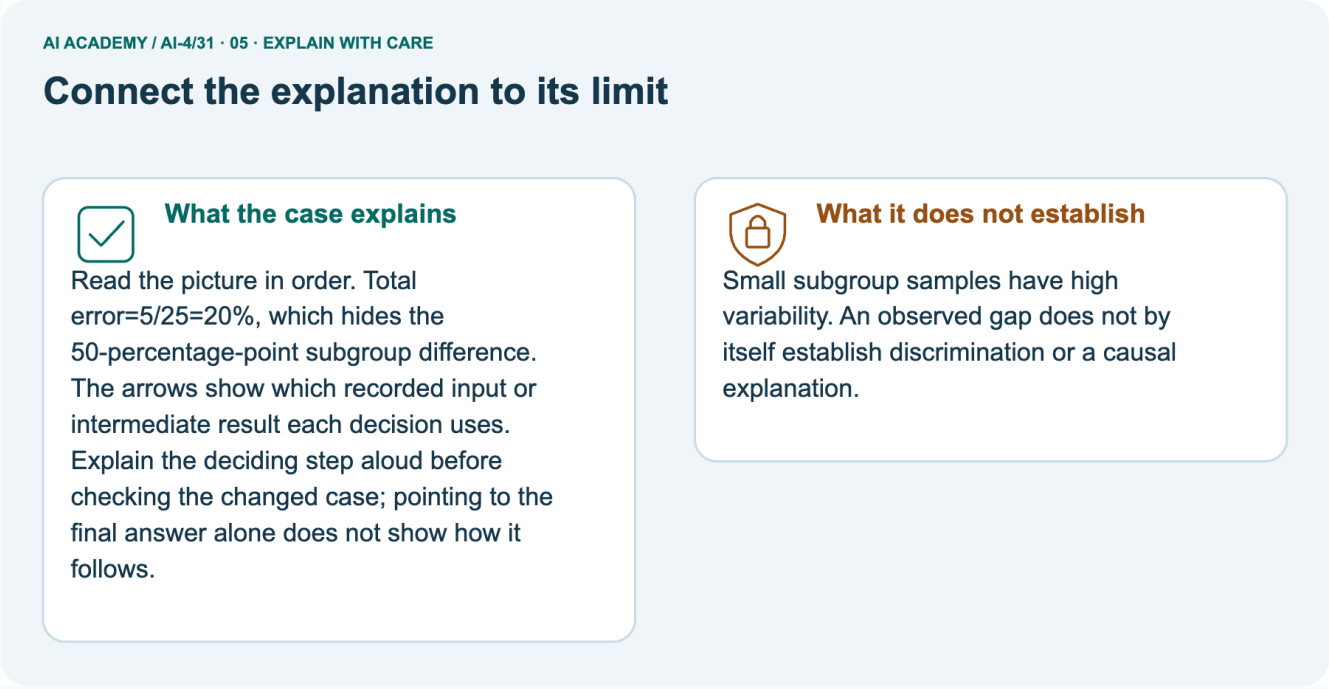
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Extra evaluated cases change **denominator**s; existing errors have not disappeared.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Small **subgroup** samples have high variability.

An observed gap does not by itself establish discrimination or a causal explanation.

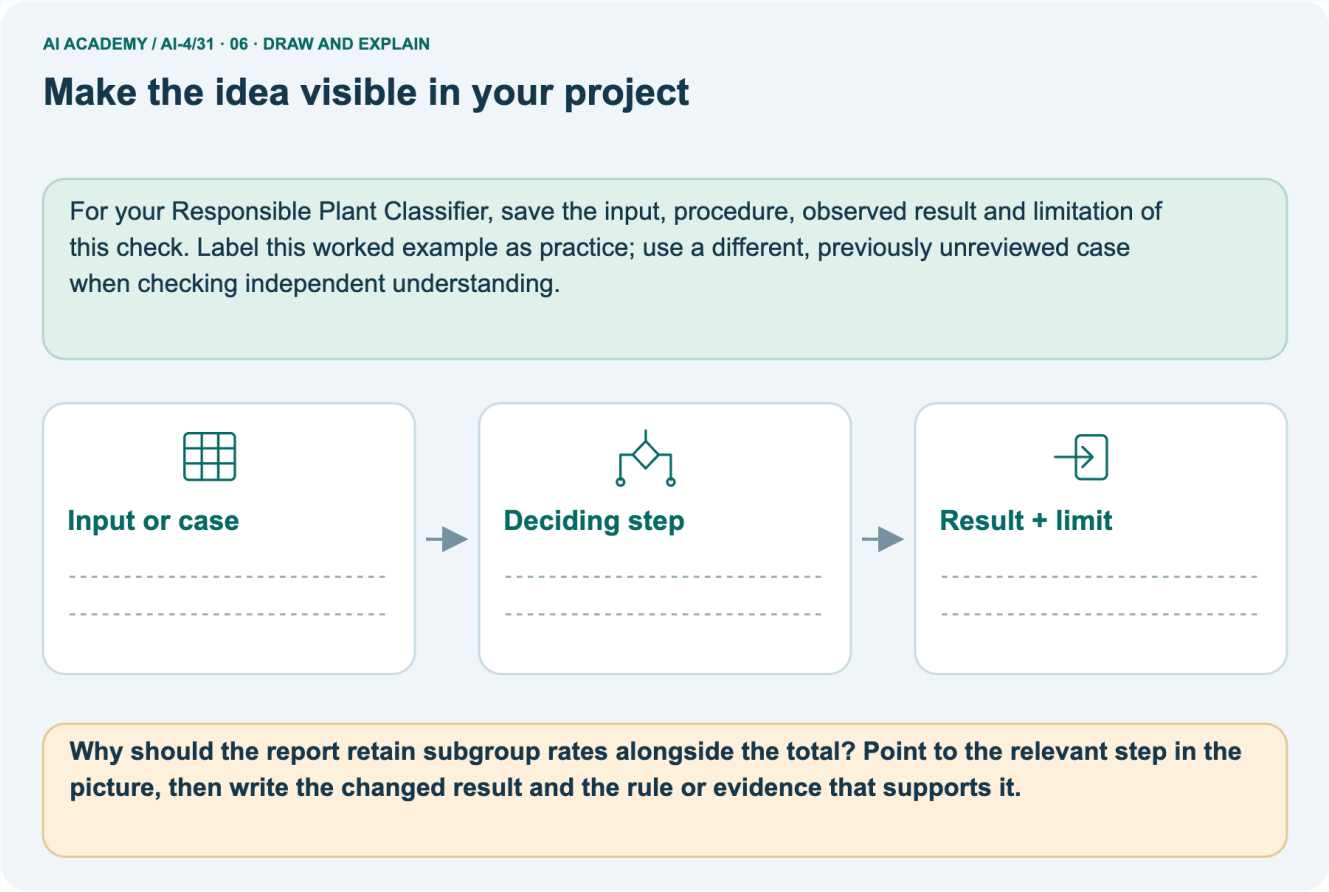
> Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Small **subgroup** samples have high variability. An observed gap does not by itself establish discrimination or a causal explanation.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the report retain **subgroup** rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The20% original total is dominated by Sun’s larger sample and understates the observed Shade difficulty. Include counts, conditions and uncertainty.

Changed-case check: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Extra evaluated cases change **denominator**s; existing errors have not disappeared.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.